Reg. No | | | | | | | | | |

Question Paper Code : 13441

B.E / B.TECH DEGREE EXAMINATION, NOVEMBER / DECEMBER 2017

Third Semester

B.E. – Computer Science and Engineering

15CS301 - DATA STRUCTURES AND ALGORITHMS

(Regulations: Mepco – R2015)

Duration: 3 Hours Max.: 100 Marks

Answer ALL Questions

PART A – (10 × 2 = 20 Marks)

1. What is an algorithm?
2. Define recurrence relation. Give an example.
3. Compare array and linked list based implementation of list.
4. Define Stack.
5. Define the degree of tree. Draw a tree with degree 3.
6. Write the heap order property.
7. Define Disjoint set.
8. Give the advantage and disadvantage of rehashing technique.
9. How will you represent graph?
10. Determine the time complexity of the Prim's algorithm.

PART B – (5 × 16 = 80 Marks)

11. a) i. Consider the following algorithm.

```
ALGORITHM Secret(A[0..n − 1])
//Input: An array A[0..n − 1] of n real numbers
minval← A[0]; maxval← A[0]
for i ← 1 to n − 1 do
   if A[i] <minval
          minval← A[i]
   if A[i] >maxval
          maxval← A[i]
return maxval–minval
```

What does this algorithm compute? What is its basic operation? Compute the time and space complexity. (10 Marks)

11. a) ii. Specify the best case, worst case and average case input for linear search. (6 Marks)

OR

11. b) i. Design an algorithm for binary search. Find the time complexity. Draw the tree of recursive calls for binary search algorithm. (10 Marks)

11. b) ii. Explain in detail about asymptotic notations. (6 Marks)

12. a) i. Imagine we have two empty stacks s1 and s2. Draw a picture of each stack after the following operations.

```
pushStack(s1,3)
pushStack(s1,5)
pushStack(s1,7)
pushStack(s1,9)
pushStack(s1,11)
pushStack(s1,13)
while(!emptyStack(s1))
{
    popStack(s1,x)
    pushStack(s2,x)
}
```

(8 Marks)

12. a) ii. Write a routine to reverse the singly linked list and print the contents. (8 Marks)

OR

12. b) i. Design a List ADT using linked list implementation.

Write the functions to perform the following operations

Find

Find Next

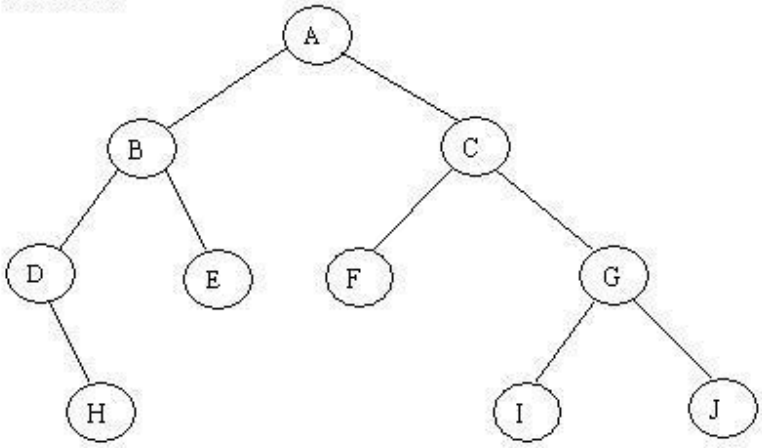
Find Previous

Count (8 Marks)

12. b) ii. Design a PrinterQueue to process the jobs for printing. (8 Marks)

13. a) i. Write the routine for in order, preorder and post order traversal. For the following tree perform the inorder, preorder and postorder traversal.

Given tree:

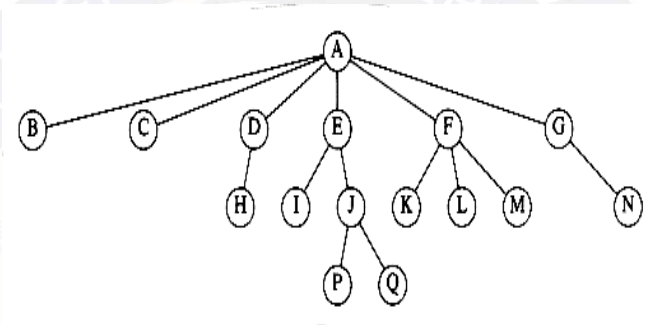


(8 Marks)

13. a) ii. Write a routine to insert an element in a min heap. Illustrate using 10, 15, 23, 7, 46, 3, 6. (8 Marks)

OR

13. b) i.



For the given tree

Find the root, leaf nodes, height and depth of a tree.

Draw the Left Child - Right Sibling representation of the given tree. (8 Marks)

13. b) ii. Why should we balance a tree? Write a routine for single rotation left and right in AVL tree. Illustrate with 1,5,7 and 20, 15, 12. (8 Marks)

14. a) i. Derive the result of the doing the sequence of operation union(1,2), union(3,4), union(3,5), union(1,5), union(1,7) in a set

A) by height

B) by size. (8 Marks)

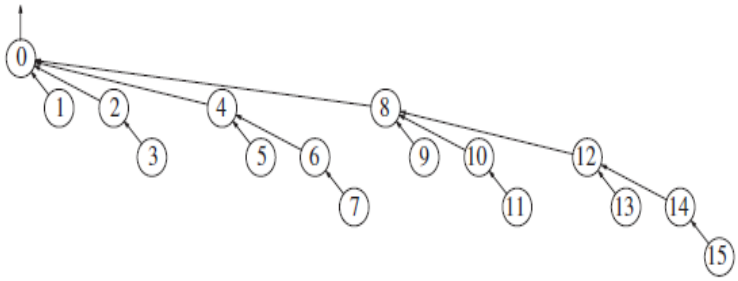
14. a) ii. Explain double hashing technique. Show the resultant hash table after inserting the elements 89, 18, 49, 58 and 69 if $hash_1(x)$ = x mod 10 and $hash_2(x)$ = 7 - (x mod 7). (8 Marks)

OR

14. b) i. Show the contents of hash table using quadratic probing for the input 4371, 1323, 6173, 4199, 4344, 9679, 1989 with a hash function x mod 10. (8 Marks)

14. b) ii. Write a function for path compression. Why we go for path compression?

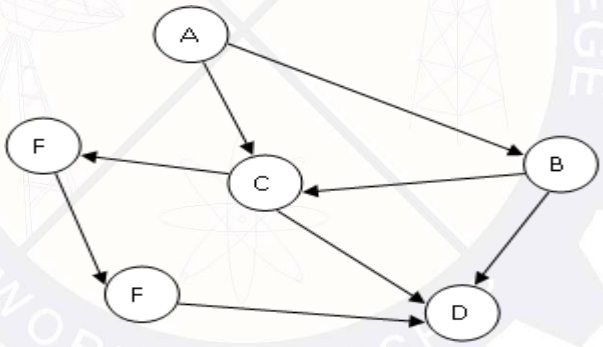
Apply this to the following tree during find(14)



(8 Marks)

15. a) i. Write and explain the pseudo code for topological sorting of a graph. Illustrate for course prerequisite graph. (10 Marks)

15. a) ii. Perform DFS traversal of a graph. Find the connected components.

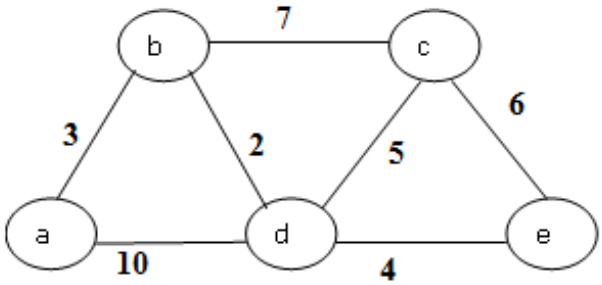


(6 Marks)

OR

15. b) i. Write and explain the single source shortest path algorithm. (10 Marks)

15. b) ii. Obtain the minimum spanning tree using Kruskal's algorithm.



(6 Marks)